The Berger–Hopf Standard Model: A Frozen v1.0 Geometric Reinterpretation of Standard Model Flavor, Couplings, and Generations

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BHSM v1.1 technical note package

Abstract

The Berger–Hopf Standard Model (BHSM) is a no-retuning, alpha-anchored geometric reinterpretation framework for Standard Model flavor, couplings, generations, and electroweak-scale structure. This technical note typesets the frozen v1.0 repository state at commit `03039feb14fb4c988edce8453f6ee5b2` and tag `bhsm-v1.0-freeze`. The frozen construction fixes the Berger anisotropy $a = \alpha^{-1}/(12\pi^2)$, the universal overlap width $S = 1/(4\pi)$, and a charged-sector mode ledger for charged leptons, up quarks, and down quarks. Two branches are kept separate: `BHSM_BARE_V1`, the pure alpha-anchored Berger–Hopf overlap model, and `BHSM_DRESSED_V1_CANDIDATE`, which applies $Z^{u,2}_{\mathrm{virt}} = 1/2$ only to c/t. The dressed branch remains a candidate, not final canonical adoption.

This note reports frozen prediction ledgers, CKM and Hopf-phase CP screens, PMNS effective-extension rows, gauge/Higgs/electroweak screens, the finite-basis H_T proxy gap audit, scalar/topographic scaffold status, fixed tolerance bands, and falsification criteria F1–F9. It does not claim a first-principles derivation of the Standard Model, an analytic proof of the full H_T no-extra-light-state theorem, or a proof of scalar decoupling from the full action.

1 Introduction

BHSM means Berger–Hopf Standard Model. It is used here as a frozen, no-retuning geometric reinterpretation framework for Standard Model flavor, couplings, generations, and electroweak-scale structure. The v1.0 freeze is not a new fit performed inside this manuscript. It is the documented repository state at commit `03039feb14fb4c988edce8453f6ee5b234797eb2`, tag `bhsm-v1.0-freeze`, where the test suite passed with 269 tests at the freeze and 278 tests on the current paper branch.

The Standard Model remains the reference low-energy quantum field theory for the gauge and matter ledger used here. BHSM is not presented as replacing the Standard Model. It is a no-retuning geometric reinterpretation framework whose outputs are compared against standard phenomenological ledgers and reference quantities.

The core claim discipline is conservative. Hypercharge and anomaly checks are ledger-level and test-backed. Flavor outputs are internal-rule screens tied to the supplied mode ledger and overlap rule. Gauge and electroweak outputs are matching screens. The H_T and scalar/topographic sectors remain proxy or scaffold audits. The dressed branch is an adoption candidate, not a final canonical model.

2 Repository and Reproducibility

This technical note is assembled from the private development repository for the Berger–Hopf Standard Model project. It is a manuscript artifact on the paper branch and does not alter the frozen model.

Item	Value
Repository	`https://github.com/ncarberry64/Berger-Hopf-Standard-Model`
Frozen tag	`bhsm-v1.0-freeze`
Model freeze commit	`03039feb14fb4c988edce8453f6ee5b234797eb2`
Manuscript branch	`bhsm-v1.1-paper`
Freeze tests	`269 passed`
Paper-branch QA tests	`278 passed`

Reproducibility command:

```
python -m pytest -q
```

3 Framework and Frozen Configuration

The v1.0 freeze fixes the alpha-anchored Berger geometry and universal overlap width before any manuscript scoring.

Table 2: Frozen constants.

Quantity	Frozen value	Status
Berger anisotropy	$a = \alpha^{-1}/(12\pi^2) = 1.157054135733433$	alpha-anchored canonical geometry
Universal overlap width	$S = 1/(4\pi) = 0.07957747154594767$	frozen overlap width

The alpha-anchored geometry is selected by the theory-side rule recorded in the model card: the BHSM scale sector contains $\epsilon_\alpha = \alpha^{-1}/(12\pi^2) - 1$. It is not selected by residual minimization.

Table 3: Fixed charged-sector mode ledger.

Sector	Heavy	Middle	Light
charged leptons	$(0, 0)$	$(5, 2)$	$(9, 3)$
up quarks	$(0, 0)$	$(6, 0)$	$(10, 1)$
down quarks	$(0, 0)$	$(6, 3)$	$(8, 2)$

The Hopf charge is $q = k - 2j$. Charged-sector ratios use the frozen overlap rule

$$\frac{m_i}{m_3} = \exp[-S\lambda_{k,j}],$$

with the Berger scalar spectrum proxy

$$\lambda_{k,j}(a) = a^2(k - 2j)^2 + 2((2j + 1)k - 2j^2).$$

4 Gauge and Field Ledger

The repository audits the Standard Model gauge and matter ledger as a conditional ledger within the admitted BHSM framework. The manuscript does not claim a derivation of the Standard Model from pure geometry alone. It records the tested statement that the admitted chiral representation pattern, Higgs-selected $U(1)$, Yukawa invariance, and anomaly cancellation reproduce the anomaly-free Standard Model ledger conditionally.

5 Flavor Prediction Ledger

5.1 `BHSM_BARE_V1` branch

Table 4: Frozen `BHSM_BARE_V1` charged-sector and CKM outputs.

Quantity	Value	Status
μ/τ	0.05985960910845228	frozen output
e/τ	0.00028830990267827904	frozen output
c/t	0.008310500554068288	scheme-sensitive comparison
u/t	$1.2690463017606151 \times 10^{-5}$	scheme-sensitive comparison
s/b	0.021933971495439474	scheme-sensitive comparison
d/b	0.0011165200546001757	scheme-sensitive comparison
$\sin\theta_{12}$	0.2256184580048353	CKM screen
$\sin\theta_{23}$	0.04386794299087895	CKM screen
$\sin\theta_{13}$	0.0035623676140463315	CKM screen

5.2 `BHSM_DRESSED_V1_CANDIDATE` branch

Table 5: Frozen `BHSM_DRESSED_V1_CANDIDATE` changes relative to `BHSM_BARE_V1`.

Quantity	`BHSM_BARE_V1`	`BHSM_DRESSED_V1_CANDIDATE`	Changed
c/t	0.008310500554068288	0.004155250277034144	yes
u/t	$1.2690463017606151 \times 10^{-5}$	$1.2690463017606151 \times 10^{-5}$	no
s/b	0.021933971495439474	0.021933971495439474	no
d/b	0.0011165200546001757	0.0011165200546001757	no
$\sin\theta_{13}$	0.0035623676140463315	0.0035623676140463315	no

The dressed branch changes only c/t using $Z^{u,2}_{\text{virt}} = 1/2$ for the pure-fiber middle up mode $(6, 0)$. It leaves u/t, CKM $\sin\theta_{13}$, down-sector ratios, lepton ratios, gauge outputs, Higgs/electroweak outputs, H_T, and scalar outputs unchanged. It remains a candidate, not final canonical adoption.

6 CKM and CP Structure

The CKM angle screens use frozen canonical BHSM ratios:

$$\sin\theta_{12} \approx \sqrt{d/s}, \qquad \sin\theta_{23} \approx 2(s/b), \qquad \sin\theta_{13} \approx \sqrt{u/t}.$$

The Hopf-phase CP screen computes

$$\delta_{\mathrm{CKM}}^{\mathrm{BHSM}} = \Delta q\sqrt{S},$$

with Δq from the light up/down Hopf charges. The frozen branch reports $\delta_{\mathrm{CKM}} = 1.1283791670955126$ and $J_{\mathrm{CKM}} = 3.1011702945437805 \times 10^{-5}$. The phase is a model-output screen, not an empirical insertion.

7 Gauge, Higgs, and Electroweak Screens

The gauge coupling screens are interpreted as electroweak-scale matching conditions:

$$\alpha_1 = \frac{1}{6\pi^2}, \qquad \alpha_2 = \frac{1}{3\pi^2}, \qquad \alpha_3 \approx \frac{7}{6\pi^2},$$

$$\sin^2\theta_W = \frac{3}{13}, \qquad \alpha_{\mathrm{EM}}^{-1} = 13\pi^2.$$

The Higgs/electroweak screen uses

$$\epsilon_\alpha = \alpha^{-1}/(12\pi^2) - 1, \qquad v = 2\sqrt{2}E_P \exp[-4\pi^2 - \epsilon_\alpha/(4\pi^2)].$$

Precision QCD/RG matching remains open; the one-loop and running scaffolds are comparison tools, not completed high-precision threshold matching.

8 H_T Gap and Scalar Sector

The no-extra-light-state condition is audited through proxy and finite-basis scaffolds. The target is

$$H_T|_{\mathcal{H}_\perp} \geq (4\pi^2 v)^2.$$

The dimensionless Hopf target is $\mu_H = 64\pi^5$. The current repository contains proxy spectral-gap, positive-semidefinite profile, Level 2 finite-basis twisted Dirac, lower-bound, and basis-convergence audits. The H_T theorem remains proxy/scaffold audited, not analytically proven.

Scalar decoupling remains scaffold audited. Scalar/topographic decoupling remains open at the full action level. The Standard-Model limit requires exactly one light Higgs projection and no unscreened light direct-coupled scalar. Full scalar decoupling from the action remains open.

9 Bare vs Dressed Branches

`BHSM_BARE_V1` is the pure alpha-anchored Berger–Hopf overlap model. `BHSM_DRESSED_V1_CANDIDATE` is the same frozen model except for the local virtual-environment factor $Z_{\mathrm{virt}}^{u,2} = 1/2$ applied only to c/t. The branches are separate ledger entries. The dressed branch is an adoption candidate, not final canonical adoption.

10 Falsification Ledger

Table 6: Frozen falsification criteria.

ID	Criterion	Status
F1	If alpha-anchored a cannot be derived from the internal action, BHSM geometry weakens.	open proof obligation
F2	If Ω_f cannot be derived from the twisted Dirac/bundle action, mass hierarchy predictions remain unsupported.	open proof obligation
F3	If scheme-consistent quark ratios disagree beyond fixed tolerance bands, BHSM flavor mapping fails or must be revised.	falsifiable numerical branch
F4	If canonical BHSM V_{us}, V_{cb}, V_{ub}, δ, and J fail outside fixed tolerances, BHSM flavor mapping is falsified or constrained.	falsifiable numerical branch
F5	If neutrino ordering, octant, or phase decisively contradict BHSM effective-extension outputs, the neutrino branch fails.	effective-extension branch
F6	If the full twisted Dirac/H_T spectrum produces extra light states below $4\pi^2 v$, the SM-equivalent BHSM mapping fails.	open spectral theorem
F7	If unscreened light scalar/topographic modes remain, BHSM fails as a Standard-Model-equivalent low-energy theory.	open action-level proof
F8	If higher-loop/threshold RG matching moves coupling agreement away from the electroweak scale beyond tolerance, the coupling branch weakens.	open RG matching
F9	Any post-freeze adjustment of a, S, modes, or Z_{virt} based on residuals invalidates the v1.0 prediction set.	freeze constraint

Table 7: Fixed tolerance bands.

Class	Tolerance
exact status	pass/fail
gauge couplings	0.01
Higgs/electroweak v	0.01
Higgs mass zeroth-order	0.02
charged lepton ratios	0.25
quark ratios, scheme-aware	0.25
quark ratios, otherwise	scheme-sensitive
CKM angles	0.10
CKM CP/Jarlskog	0.10
PMNS effective	0.05
H_T gap	binary pass/fail
scalar decoupling	binary scaffold pass/fail

11 Limitations and Open Proof Obligations

Table 8: Open proof obligations.

Topic	Current limitation
H_T spectrum	Proxy/scaffold audited, not analytically proven; full twisted Dirac spectrum remains open.
Scalar decoupling	Scaffold audited; full action-level scalar/topographic decoupling remains open.
Boundary operators Ω_f	Action-linked, not fully action-derived from the twisted Dirac/bundle action.
Precision QCD/RG matching	Higher-loop, threshold-aware, scheme-consistent matching remains open.
Dressed branch	Candidate only; not final canonical adoption.
PMNS sector	Effective-extension screen, not a minimal Standard Model prediction.

The no-retuning rule remains central: changing a, S, the supplied mode ledger, tolerance bands, or Z_{virt} based on residuals invalidates the frozen package.

12 Publication Declarations

Funding. No external funding is declared unless otherwise specified by the author.

Conflicts of interest. None declared unless otherwise specified by the author.

Data availability. No new observational data are introduced. Code and repository availability are by author decision.

Code availability. The BHSM v1.0 freeze is identified by tag `bhsm-v1.0-freeze` and commit `03039feb14fb4c988edce8453f6ee5b234797eb2`.

AI/tool assistance. Manuscript and repository organization used AI-assisted drafting/coding support; scientific responsibility remains with the author.

13 Conclusion

The frozen BHSM v1.0 repository defines a reproducible no-retuning prediction and falsification package for a Berger–Hopf reinterpretation of Standard Model flavor, couplings, generations, and electroweak-scale structure. This v1.1 paper branch turns that package into a submission-oriented technical note without changing the frozen model. The next technical work remains action-level derivation of Ω_f, full analytic H_T spectral computation, scalar/topographic decoupling from the full action, and precision QCD/RG matching.

14 References

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